

Tracing Functional Correctness in DLMs with Sparse Autoencoders

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Abstract

Prior probing of diffusion language models (DLMs) suggests that correctness signal grows monotonically through denoising. Using DLM-Scope sparse autoencoders, we instead trace the signal feature by feature across two DLMs and four tasks. Sparse fail-vs-pass concentration appears in task-dependent phases: code and structural tasks form a broad mid-denoising plateau (LLaDA-MBPP, Fisher $p < 10^{-7}$), whereas reasoning tasks emerge only at the final state, with cross-model agreement on 3/4 tasks. Two persistent features hand off within the late plateau, marking an internal transition rather than feature swapping. Single- and multi-feature steering across 202 LLaDA-MBPP samples yields 0/202 label flips within the plateau (step 64) and only 3/202 before plateau formation (step 16): the sparse features behave as diagnostic markers, not controllable knobs. To probe this null we compare per-layer hidden-state correctness encoding across Dream-7B-Base, its autoregressive base Qwen-2.5-7B-Base, and Qwen-2.5-7B-Instruct on the same prompts ($n=257$). All three reach equivalent peak 5-fold CV-AUC (0.71–0.73) but at different layers (Qwen-Base L19, Qwen-Instruct L20, Dream-Base L26): diffusion training shifts the locus of correctness encoding without expanding its capacity over instruction tuning. A direct test sharpens the reading: the Dream L26 probe predicts Qwen-Instruct’s pass/fail at AUC 0.68, essentially matching its 0.72 on Dream’s own labels, so the L26 representation tracks *which prompts are hard* rather than *which generations are wrong*. The LLaDA L26 SAE plateau is most consistent with the same difficulty interpretation, which accounts for the steering null without invoking a generator-specific mechanism.

1 Introduction

Diffusion language models (DLMs) generate text by iteratively denoising a fully masked sequence over ~ 100 steps (Austin et al., 2021a; Sahoo et al.,

2024; Lou et al., 2024; Nie et al., 2025; Ye et al., 2025). Chiu (2026) trains scalar probes on hidden states and reports that aggregate correctness AUC accumulates *monotonically* through denoising, with a 0.08–0.11 gain from step 0 to the final state. That scalar view leaves three questions open: (i) *which* sparse representational components carry the failure signal, (ii) *when* during denoising they form, and (iii) whether the discovered components are merely diagnostic or also causally control correctness. Decomposing the same residual stream into named SAE features rather than a single scalar score lets us answer all three, and uncovers temporal structure that the monotonic AUC curve hides.

Contributions.

- **Temporal anatomy of sparse failure signal.** On LLaDA-8B / MBPP, a 3-seed dense sweep reveals a mid-denoising plateau over steps 48–116 with a within-plateau hand-off from f15601 to f3892 (Fisher $p < 10^{-7}$); peak phase agrees across LLaDA-8B and Dream-7B on 3/4 tasks.
- **Steering null at scale.** Across 202 LLaDA-MBPP samples and 23 linear-intervention protocols ($n=345$ in total), the three within-plateau conditions are null at 0/202 each; only the pre-plateau (*s16*) intervention produces a marginal 3/202 $\approx 1.5\%$ effect.
- **Diffusion shifts the locus, not capacity.** A three-way per-layer comparison across Dream-7B-Base, Qwen-2.5-7B-Base, and Qwen-2.5-7B-Instruct shows the same peak CV-AUC (~ 0.72) at different layers (L19, L20, L26 respectively); a cross-generator test (Dream L26 probe predicts Qwen-Instruct correctness at AUC 0.68, matching its 0.72 on Dream labels) frames the L26 representation as encoding *problem difficulty* rather

than *generation-specific correctness*, consistent with the steering null.

Code and figures will be released on acceptance.

2 Related Work

Discrete diffusion and masked DLMs generate text by iteratively denoising masked tokens rather than continuing left to right (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024; Nie et al., 2025; Ye et al., 2025). Prior probing of the same models and tasks studies aggregate hidden-state predictability and reports monotonic gains in correctness AUC through denoising. SAEs decompose residual streams into sparse named features and have been used primarily to interpret autoregressive LMs (Bricken et al., 2023; Cunningham et al., 2024; Gao et al., 2024). DLM-Scope extends this toolkit to DLMs with Top- K SAEs (Makhzani and Frey, 2014; Gao et al., 2024) trained on masked-position residuals across mask rates $t \sim U(0, 1)$ (*Mask-SAE*), keeping the full denoising trajectory in distribution (Wang et al., 2026). Recent AR SAE studies report both successful steering for refusal and instruction following (O’Brien et al., 2024; He et al., 2025) and cautions that contrastive features may track lexical correlates or that steering degrades unrelated capabilities (Galichin et al., 2025; Korznikov et al., 2025); Tahimic and Cheng (2025) target code-correctness signals with SAEs on AR LMs, closest to our MBPP setting but without temporal trajectories. Goel et al. (2026) contrast representation evolution between diffusion and AR LMs, motivating our cross-DLM step-axis comparison. We reuse the DLM-Scope SAEs to ask whether the scalar correctness trend on DLMs decomposes into temporally localised sparse features, treating the steering question as open rather than presumed causal.

3 Method

Models and SAEs. LLaDA-8B (Nie et al., 2025) (block-wise unmasking, block length 32) and Dream-7B (Ye et al., 2025) (global linear schedule) both denoise a ≤ 256 -token generation region over $T=128$ steps conditioned on a fully visible prompt. We use the DLM-Scope L26 LLaDA-mask SAE and L23 Dream-mask SAE (trainer index 2, $d_{\text{sae}}=16,384$, $K=160$). The AR comparison (DLM-Scope L23 SAE on Qwen-2.5-7B (Team, 2024)) is a single-token reference point, not a trajectory: we encode the last-prompt-token resid-

ual once per sample as a single-step analog of the DLM’s pre-generation state.

Three-way per-layer AR-DLM scan. Dream-7B is architecturally identical to Qwen-2.5-7B (28 layers, $d_{\text{model}}=3584$, identical attention head counts and vocab), enabling layer-aligned comparison. Throughout the paper, “Dream-Instruct” denotes the checkpoint used for all DLM generation in the cross-model cells; “Dream-Base” denotes the sibling diffusion checkpoint without instruction tuning, used only as the hidden-state source for the AR-DLM scan. For all 257 MBPP-sanitized prompts we collect the last-prompt-token hidden state at every transformer block ($L \in \{0, \dots, 27\}$) for three models under the same chat-template prompt: Qwen-2.5-7B-Base (AR pretraining), Qwen-2.5-7B-Instruct (AR pretraining + instruction tuning), and Dream-7B-Base (AR pretraining + masked diffusion fine-tuning from the same Qwen base). We then fit per-layer logistic probes (5-fold CV, $C=0.01$) on the Dream-Instruct fail/pass labels (so all three models predict the same generation outcomes) and report CV-AUC. This isolates two effects against the shared AR base: instruction tuning (Qwen-Base $\rightarrow$ Qwen-Instruct) and diffusion fine-tuning (Qwen-Base $\rightarrow$ Dream-Base).

Data. We reuse cached hidden states and functional-correctness labels from Chiu (2026): MBPP-sanitized (257) (Austin et al., 2021b), JSON schema (272), GSM8K (1,319) (Cobbe et al., 2021), and ARC-Challenge (1,172) (Clark et al., 2018), generated with seed 0 and $T=128$ steps. Hidden states are mean-pooled over four equal-length generation regions at seven checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$.

Per-step SAE encoding. At each (model, step, layer) cell we take the region-mean hidden state $h \in \mathbb{R}^d$ per sample and encode it: $z = \text{TopK}(\text{ReLU}(W_{\text{enc}}(h - b_{\text{dec}}) + b_{\text{enc}})) \in \mathbb{R}^{d_{\text{sae}}}$, with $W_{\text{enc}} \in \mathbb{R}^{d_{\text{sae}} \times d}$, $W_{\text{dec}} \in \mathbb{R}^{d \times d_{\text{sae}}}$, $b_{\text{enc}} \in \mathbb{R}^{d_{\text{sae}}}$, $b_{\text{dec}} \in \mathbb{R}^d$, and TopK keeps the $K=160$ largest entries. Feature i is *active* on a sample iff $z_i > 0$.

Fail-vs-pass enrichment. Let $\mathcal{F}, \mathcal{P}$ be the fail and pass sets at a cell. Feature i ’s enrichment is the difference of active rates, $\text{enr}(i) = \frac{1}{|\mathcal{F}|} \sum_{s \in \mathcal{F}} \mathbb{1}[z_i^{(s)} > 0] - \frac{1}{|\mathcal{P}|} \sum_{s \in \mathcal{P}} \mathbb{1}[z_i^{(s)} > 0]$. We use the top-20 most fail-enriched features as the analysis subset per cell.

Cluster discriminability with permutation null.

We run KMeans on the fail samples for $K \in \{2, 3, 4, 5\}$ using the top-20 features (continuous activations) and report the silhouette of the best K . The permutation null shuffles labels $1000\times$, re-selects the top-20 features on each shuffled label set, then recomputes silhouette, controlling for the post-hoc feature selection. We report $p = \Pr[\text{silhouette}_{\text{perm}} \geq \text{silhouette}_{\text{obs}}]$. The silhouette is computed on fails only and measures sub-structure within failures, not fail-vs-pass separability directly. We therefore interpret it as *concentration of failure modes* in a selected sparse subspace, and cross-check fail/pass separability with supervised AUC in the same subspace under per-fold selection (Appendix E).

Steering. We write *fail*→*pass* for a sample whose label flips from fail to pass after intervention, and *pass*→*fail* for the reverse pass-side regression. Given a target feature i^* and strength α , a forward hook on the SAE layer’s transformer block re-encodes the residual at every step $t \geq t_0$, reads z_{i^*} , and in-place subtracts $\alpha \cdot z_{i^*} \cdot W_{\text{dec}}[:, i^*]$. $\alpha < 0$ amplifies the feature (reverse intervention); for multi-feature steering we sum contributions over a set of indices. A zero-out smoke test (force $h \leftarrow 0$) confirms the hook propagates. It garbles trailing tokens but leaves the early-committed function body intact, consistent with LLaDA committing blocks 0–3 by step ~ 48 .

4 Results

Mid-denoising plateau. Figure 1 reports a 21-step trajectory for LLaDA-MBPP across three seeds: anchors $\{0, 1, 4, 16, 32\}$, early plateau $\{48, \dots, 80\}$, late plateau $\{84, \dots, 124\}$, and final state 127. Rather than a sharp peak at step 64, the trajectory shows a broad plateau over steps 48–116: the signal-to-null gap stays in $[+0.19, +0.41]$ and only collapses at step 124. f15601 is the top-1 fail-enriched feature at 7/9 early-plateau dense steps and remains dominant through step 100; f3892 replaces it at steps 108–116, indicating a within-plateau hand-off rather than feature swapping outside the plateau. At steps 0–16 the top fail feature instead changes every step (Appendix A), consistent with the residual stream not yet encoding committed-token decisions when most positions are still masked. Figure 3 contrasts the per-sub-step gap on LLaDA-MBPP and Dream-MBPP: LLaDA-MBPP has 6–9 of 14 dense sub-steps at $p < 0.05$

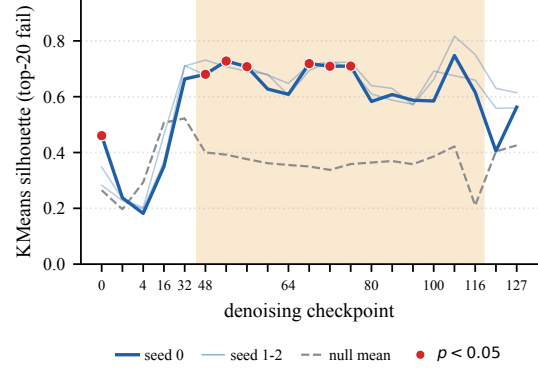


Figure 1: LLaDA-8B / MBPP: KMeans silhouette on the top-20 fail-enriched features vs. permutation null mean across denoising steps. All 21 checkpoints (anchors $\{0, 1, 4, 16, 32\}$, early plateau $\{48-80\}$, late plateau $\{84-124\}$, final 127) are from a single new-prompt generation run; faint lines overlay two additional generation seeds (Appendix I). Shaded region marks the mid-denoising plateau $[48, 116]$ where most steps reach $p < 0.05$; step 64 sits inside the plateau as a local dip, not a sharp peak.

per seed (Fisher combined $p = 5.99 \times 10^{-10}$ for seed 0), while Dream-MBPP shows the same plateau shape with a much lower concentration (gap never exceeds $+0.16$, no sub-step reaches $p < 0.05$). The LLaDA $\times$ MBPP combination is therefore a model-specific point where the residual stream factors fail-vs-pass into a small set of sparse features rather than a generic property of code generation. The raw-probe AUC in Chiu (2026) monotonically increases over the same steps (LLaDA-MBPP, ~ 0.75 at s_0 to ~ 0.84 at s_{64}), framing denoising as gradual scalar sharpening, whereas the feature-level plateau and the late hand-off are invisible to scalar probing.

Cross-model $\times$ cross-task phase agreement.

Figure 2 reports the full 2×4 trajectory grid; Appendix C lists step-64 numbers. Code and structural cells (MBPP, JSON on both DLMs) peak mid-denoising at step 64, reasoning cells (GSM8K on both, Dream-ARC) peak at step 127, and LLaDA-ARC is an outlier peaking at step 4. Three of four tasks therefore share a peak step across LLaDA and Dream, mirroring the structural-vs-reasoning emergence split of Chiu (2026) as a peak position rather than a continuous AUC curve. No cell survives Holm-Bonferroni across 40 sparse-grid tests (8 cells $\times$ steps $\{4, 16, 32, 64, 127\}$, dropping the two fully-masked steps $\{0, 1\}$ where the null is dominated by mask geometry), but two cells reach aggregated significance under Fisher’s com-

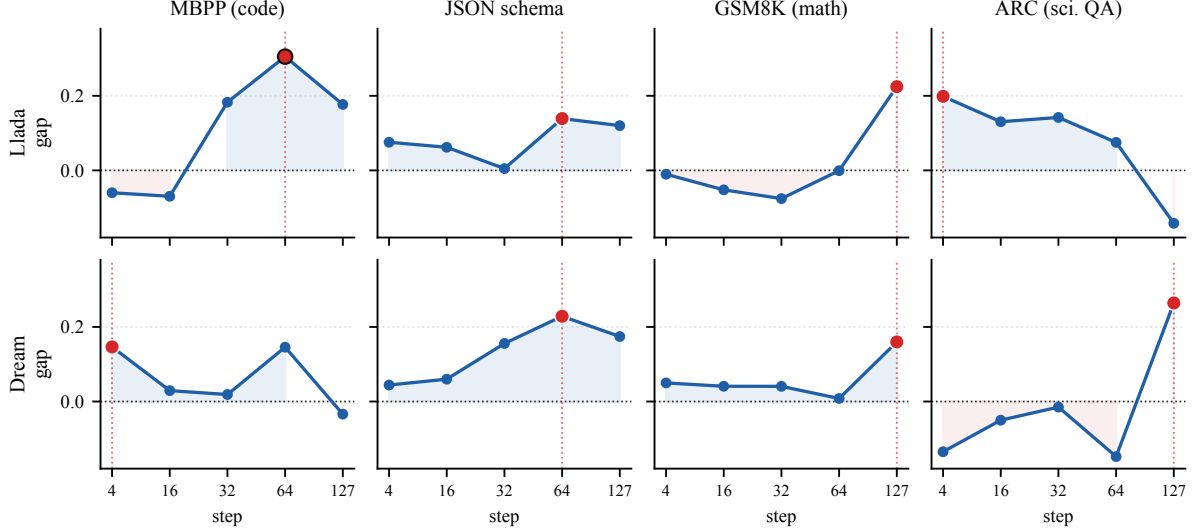


Figure 2: Cross-model $\times$ cross-task trajectory grid. Each panel plots signal-to-null gap (observed KMeans silhouette minus permutation null mean) across denoising steps for one (model, task) cell; red marker = peak gap step; black outline = $p < 0.05$ permutation significance. Code (MBPP) and structural (JSON) tasks peak at mid-denoising (step 64) on both DLMs. Math (GSM8K), and on Dream science QA (ARC), peak at late denoising (step 127). The peak step shows task-level agreement across LLaDA-8B and Dream-7B on three of four tasks; only ARC disagrees (LLaDA peak step 4, Dream step 127).

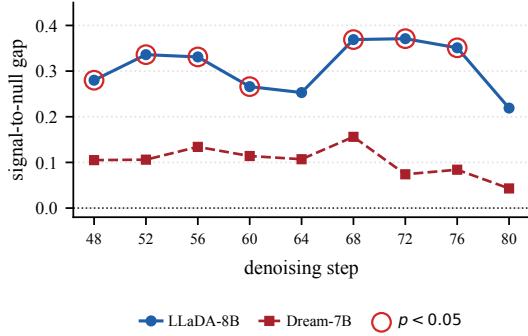


Figure 3: Dense step sweep (48–80 every 4 steps) on LLaDA-MBPP and Dream-MBPP, seed 0. Hollow red rings mark LLaDA sub-steps with permutation $p < 0.05$ (7 of 9); no Dream sub-step reaches the same threshold under the matched null. The mid-denoising plateau on LLaDA-MBPP aggregates to Fisher $p < 10^{-7}$ on every seed.

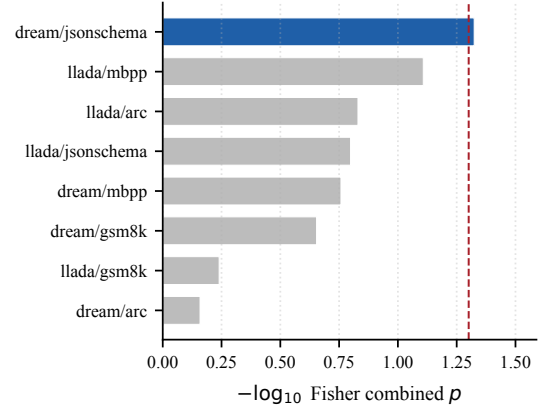


Figure 4: Fisher’s combined p across the sparse 5-step grid per (DLM, task) cell; red dashed line at $p = 0.05$. LLaDA-MBPP and Dream-JSON reach aggregated significance; reasoning tasks do not, and LLaDA-ARC remains an outlier driven by prompt-boundary failure modes.

bin probability test on each cell’s sparse 5-step trajectory (Figure 4): the LLaDA-MBPP dense plateau (Fisher $p < 10^{-7}$ per seed) and Dream-JSON ($p = 0.047$); the LLaDA-MBPP plateau is also L26-specific under a cross-layer scan (Appendices F, G, H). The ARC anomaly fits this picture: LLaDA-ARC fails near the prompt boundary so its early-step cluster captures prompt-attention failure modes rather than reasoning-stage failure, accounting for the step-4 peak versus Dream-ARC’s step-127 peak.

Top-feature drift and persistence. On LLaDA-8B / MBPP, pairwise Jaccard of the top-20 sets

is high among steps 0–4 (≥ 0.48), moderate between 32 and 64 (0.29, the plateau regime), and falls to 0.14 between 64 and 127. The dense sweep (Appendix F) ranks f15601 top-1 at 7 of 9 dense steps in [48, 80], supporting a persistent statistical marker inside the plateau and a concentration effect in sparse feature space rather than a monotonic increase in correctness information. A supervised cross-check confirms this picture: top-20 SAE features recover 85–95% of the raw-probe AUC at step 64, with raw probing winning in 10 of 12 cells (the two SAE wins are near-chance cells), so the

condition	from	fail→pass	pass→fail
suppress f15601	s64	0/152	0/50
suppress f15601	s16	2/152	1/50
suppress top-5	s64	0/152	0/50
reverse f15601	s64	0/152	0/50

Table 1: Full steering intervention summary on LLaDA-8B / MBPP; entries show flips / evaluated cases over all 152 fail samples (126 cluster-1 + 26 cluster-0) and 50 pass-side regression controls. Suppress f15601 or the top-5 features at step 64 (within the plateau) is fully null at 0/202; suppress f15601 from step 16 (before plateau formation) produces a small 3/202 effect (1.5%, both fail→pass conversions occur on cluster-1 cases). Reverse intervention ($\alpha = -5$) is null.

feature	dominant context (F/P)	top quotation
f15601	complex-math string (25/5)	doc- “nth octagonal number...”
f3892	:param/:return (15/15)	“:param tuple_list: list”
f11265	broken-syntax / incomplete (17/13)	“dp[n == 0] = [1 \n”

Table 2: Activation-maximising contexts (top-30 (*sample, position*) entries per feature) on LLaDA-8B / MBPP, step 64. f15601 concentrates on complex-math docstrings, fail/pass split 25/5. f11265 fires on already broken syntax. Full table including f5561, f2144 in Appendix L.

sparse subspace captures most yet not all of the dense-residual signal (Appendix E, Figure 9).

Steering result. We test whether the cluster-level signature can be directly controlled on LLaDA-MBPP, the only cell with cluster significance. The steering generation is a separate seed-0 decoding pass with 105/152 pass/fail (versus the dense sweep’s 115/142, Appendix I); we report all 152 fails and 50 pass-side regression controls across four interventions (Table 1). Suppressing f15601 or the top-5 fail features within the plateau (step 64), and the reverse intervention ($\alpha = -5$), each yield 0/202 label flips, as does a counterfactual replacement that subtracts the top-5 fail-enriched and adds the top-5 pass-enriched directions. A broader sweep of 23 linear protocols on small n (Appendix J) covering both sparse SAE directions (17 configs, magnitudes up to $\alpha = 50$) and raw residual directions (mean-difference, PCA, Fisher LDA; 6 configs) yields 0/345 fail→pass total. Only step-16 intervention, before plateau formation, produces a marginal effect (3/202). A zero-out smoke test ($h \leftarrow 0$) confirms the hook propagates: it garbles trailing tokens but leaves the early-committed function body intact, consistent with LLaDA committing blocks 0–3 by step ~ 48 .

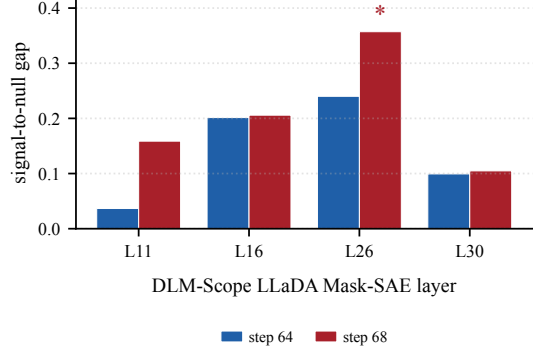


Figure 5: LLaDA-MBPP signal-to-null gap across SAE layers at plateau steps 64 and 68. L26 is the unique layer where the gap reaches permutation $p < 0.05$ (step 68, gap +0.357). Top-1 fail-enriched feature differs by layer; f15601 appears only at L26.

What f15601 actually encodes. To turn the statistical marker into an interpretable variable we re-run LLaDA at step 64 on all 257 MBPP samples, encode every generation position through the L26 Mask-SAE, and record the top-30 (*sample, position*) entries per target feature. Table 2 shows the result for three top features (full table in Appendix L). The mid-plateau peak feature f15601 concentrates on docstring positions containing dense mathematical specifications (Eulerian / octagonal / hexagonal / Babylonian formulae): in its top-30 activations the fail/pass split is 25/5, but the context is the *problem statement* rather than any concrete failure mode in the model’s output. This makes f15601 a *task-difficulty correlate* that fires upstream of any specific failure: ablating it removes the model’s encoding of “this problem is hard,” which is necessary for predicting failure but not sufficient to change the generation trajectory. The late-plateau replacement f11265 fires on contexts that already contain visible syntactic defects in the partially-generated code (top quotation “dp[n == 0] = [1 \n dp = [0]]”), suggesting the within-plateau hand-off from f15601 to f3892 (and the late drift toward f11265) reflects a transition from *difficulty signalling* to *outcome signalling* as denoising completes.

L26 is the unique cluster-significant layer. To check that L26 is not an arbitrary choice, we re-run the silhouette-with-permutation-null pipeline at the four available DLM-Scope LLaDA SAE layers {11, 16, 26, 30} at plateau steps 64 and 68 (Figure 5). L26 is the unique peak: step 68 at L26 gives gap +0.357 with permutation $p = 0.006$, while L11, L16, and L30 give gaps below +0.21 and none reach $p < 0.05$. Moreover f15601 appears as

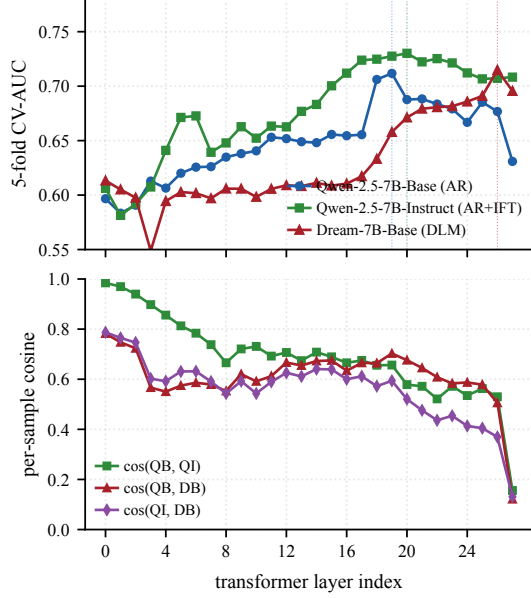


Figure 6: Per-layer 5-fold CV-AUC for predicting Dream-Instruct MBPP correctness from the last-prompt-token residual at each transformer block (top), and per-sample cosine similarity between the three models (bottom). All three models are architecturally identical (28 layers, $d=3584$, same vocab). Qwen-Base peaks at L19 (AUC 0.71), Qwen-Instruct at L20 (0.73), Dream-Base at L26 (0.72); peak capacity is shared but the locus shifts. At L27, both post-trained models diverge sharply from Qwen-Base ($\cos < 0.16$) and from each other ($\cos 0.13$), indicating post-training rewrites the final layer regardless of objective.

model	peak L	peak AUC	L27 AUC	$\cos_{27}$
Qwen-Base (AR)	19	0.712	0.631	1.00
Qwen-Instruct	20	0.730	0.708	0.16
Dream-Base (DLM)	26	0.715	0.696	0.12

Table 3: Per-layer CV-AUC peaks on MBPP-Dream labels ($n=257$). Peak AUC is similar but the locus shifts from L19/20 to L26. At L27 both post-trained models diverge from Qwen-Base ($\cos < 0.17$); the rewrite is not diffusion-specific.

top-1 fail-enriched feature only at L26: different layers select different atoms (f13087/f541 at L11, f2741/f15654 at L16, f8020/f654 at L30). The persistent-feature claim is therefore L26-specific rather than a generic mid-layer property, supporting our choice of L26 as the analysis layer. Repeating the suppression protocol of §4 at the three non-L26 layers using each layer’s top-1 fail feature ($\alpha = 5$, $t \geq 64$, $n = 60$ per layer) yields 0/60 flips at each layer: the within-plateau steering null generalises across the layer span tested, so the difficulty-encoding interpretation is not L26-specific (Appendix G).

Diffusion shifts the locus of correctness encoding without expanding capacity. Figure 6 shows the per-layer CV-AUC for the three architecturally matched 7B models on the same prompts and Dream-Instruct labels ($n=257$); Table 3 summarises the three peaks. Three observations: (i) the three models reach equivalent peak AUC (0.712, 0.730, 0.715) but at different layers (L_{19} , L_{20} , L_{26}); (ii) diffusion training does not exceed instruction tuning at peak (Qwen-Instruct’s L20 peak is the highest of the three); (iii) post-training rewrites the final layer regardless of objective: at L27 both Qwen-Instruct and Dream-Base diverge from Qwen-Base ($\cos 0.16$ and 0.12 respectively) and from each other ($\cos 0.13$). The L27 collapse of Qwen-Base AUC (from 0.71 at L19 to 0.63) is recovered by both post-training procedures (0.71 for Qwen-Instruct, 0.70 for Dream-Base), so the late-layer divergence is not diffusion-specific. Critically, Qwen-Instruct does not generate the Dream solutions, yet matches Dream’s L26 peak with an L20 peak of comparable AUC: this implies the per-layer AUC reflects *problem-difficulty encoding*, which is shared across architectures and post-training procedures, rather than DLM-specific generation-correctness encoding.

5 Discussion

Why linear steering fails. The three-way scan supplies a candidate mechanism: Dream’s L26 representation encodes *which prompts are hard*, not *which generations are wrong*, and the same difficulty interpretation is the most parsimonious account of LLaDA’s L26 SAE plateau, which is what steering targets. A linear suppression at L26 therefore subtracts a difficulty estimate from the residual; the underlying generation trajectory, decided by token-level decoding decisions throughout denoising, is unaffected.

Cross-generator test. We test the difficulty interpretation directly by re-running MBPP-sanitized with Qwen-2.5-7B-Instruct (pass rate $190/257 = 74\%$, per-sample pass/fail agreement with Dream-Instruct 0.642) and asking how well Dream’s L26 hidden states predict Qwen-Instruct’s correctness on the same problems. Under 5-fold logistic CV with $C = 0.01$, the Dream L26 probe predicts Qwen-Instruct labels at AUC 0.681, essentially matching its 0.715 on Dream’s own labels (diff -0.033). The L26 probe therefore transports across generators with little AUC loss, suggesting

much of the predictive signal is shared problem difficulty rather than Dream-specific self-correctness. This reading aligns with the three steering signatures: the full 0/345 flip rate across sparse and raw-residual directions, the marginal 3/202 effect only at step 16 before any token commits, and Qwen-Instruct’s higher L20 AUC despite not being the generating model. A capacity-limited DLM control protocol therefore needs features that encode commitments rather than difficulty, plausibly at earlier layers or denoising steps, or non-linear interventions that re-route the trajectory rather than scaling a single direction.

Relation to AR SAE steering successes. Recent AR SAE work reports successful steering for refusal (O’Brien et al., 2024) and instruction following (He et al., 2025); Tahimic and Cheng (2025) steer code-correctness signals at the SAE layer on AR LMs. The difficulty-encoding mechanism we identify is consistent with that body of work rather than in tension with it: refusal and instruction following are stylistic axes that are not predictable from problem difficulty, so an AR SAE feature can causally affect them without overlapping with a difficulty signal. Code correctness is the harder target: Tahimic and Cheng (2025) achieves a smaller-than-refusal effect size, and our DLM cross-generator test suggests that what AR correctness probes detect is partly a difficulty correlate as well. We therefore read the DLM steering null as a sharpened instance of a general SAE-correctness limitation rather than a DLM-specific failure mode; the per-layer scan further suggests that for DLMs the steering layer should track the model’s self-correctness peak rather than the SAE plateau.

Future directions. Three concrete avenues follow from the present results. *Commitment-level features.* The actmax contexts show f11265 as the late-plateau feature that fires on *already broken* partial code (Table 2); a Mask-SAE trained on Unmask positions (already revealed tokens) and selected on the late hand-off step might surface a similar outcome-side feature whose ablation is closer to causal. *Multi-layer joint steering.* The cross-layer specificity test (Figure 5) and the AR-DLM per-layer scan (Figure 6) both indicate that the correctness-related representation is layer-distributed; a single-layer linear intervention cannot recover the multi-layer write that distinguishes Dream from Qwen-Instruct. *Pre-commit interventions.* The marginal 3/202 flips occur only when

the intervention starts at step 16, i.e. before any block commits; targeting earlier steps with appropriately matched directions is the most direct way to test the commitment-timing constraint.

6 Conclusion

Tracing functional-correctness signal feature by feature reveals temporal structure that scalar probes miss: sparse failure markers form task-dependent phases, with code and structural tasks on a mid-denoising plateau and reasoning tasks peaking at the final state (Fisher $p < 10^{-7}$ on LLaDA-MBPP, cross-model agreement on 3/4 tasks). A 3-seed dense sweep confirms the steps 48–116 plateau and a late f15601→f3892 hand-off; top-20 sparse features recover 85–95% of the raw-probe AUC. Linear interventions across 202 LLaDA-MBPP samples flip 0/202 within the plateau and 3/202 before plateau formation. A three-way per-layer scan of Dream-7B-Base and its AR siblings Qwen-Base and Qwen-Instruct supplies a candidate mechanism: all three reach equivalent peak CV-AUC (0.71–0.73) but at different layers, and Qwen-Instruct (which does not generate the Dream solutions) matches the Dream L26 peak, so the L26 representation tracks problem difficulty rather than generation-specific correctness. The LLaDA L26 SAE plateau is most consistent with the same difficulty signal, accounting for the steering null; causal control therefore needs commitment-level features at earlier layers or steps, not amplification of the difficulty signal already inherited from the AR base.

Limitations

SAE coverage. All findings depend on a single DLM-Scope Mask-SAE per model (L26 for LLaDA, L23 for Dream), trained with mask rate $t \sim U(0, 1)$ on masked-position residuals. The full denoising trajectory is therefore in distribution, but features extracted at other layers, the Unmask-SAE trained on already-revealed positions, or higher-capacity dictionaries ($K > 160$) may surface different failure correlates. Early-step instability (steps 0–16) reflects the residual stream not yet encoding committed-token decisions rather than SAE distribution shift.

Feature semantics. We identify persistent dictionary atoms such as f15601 and f3892 by statistical enrichment and temporal persistence, not by human-validated semantic labels. The present experiments therefore support a claim

about when sparse failure markers concentrate; Appendix L characterises four of them via activation-maximising contexts, but full mechanistic naming would require systematic intervention beyond this protocol.

Permutation test interpretation. Our permutation null re-selects the top-20 fail features on each shuffled label set before recomputing silhouette. This is conservative (any post-hoc selection procedure that could find clusters on random labels is included in the null), and it inflates the null mean noticeably (e.g. 0.36 for LLaDA-8B / MBPP at step 64). A consequence is that only two of twelve cells reach $p < 0.05$ at an individual step on the sparse 7-grid even though the DLM grid shows systematic plateau structure. Aggregated tests partially mitigate this: the 14 dense LLaDA-MBPP plateau sub-steps yield Fisher combined $p < 10^{-7}$ on every seed and held-out feature selection (Appendix K) reaches $p < 0.05$ on all 9 dense sub-steps; we present cross-model claims as plateau-shape claims rather than per-cell significance.

Steering coverage. Our intervention sweep tests four denoising windows, two feature-set sizes (single and top-5), and one magnitude pair ($\alpha \in \{+5, -5\}$) at a single SAE layer per DLM (L26 for LLaDA, L23 for Dream). We do not test multi-layer joint interventions, non-linear projections, or feature-selection strategies that use the per-step top-1 (where the top fail feature differs from f15601). Effect-size diagnostics (Appendix J) confirm the intervention reaches $\|\Delta h\|/\|h\| \in [4.4\%, 14.8\%]$ at the SAE-layer residual, so the within-plateau null is not because the perturbation was silent; rather, a 0/202 result on this protocol is narrower than a negative result on all possible linear-feature interventions.

Cluster K and choice of metric. We use silhouette on KMeans with $K \in \{2, 3, 4, 5\}$ and report the best K per cell. On LLaDA-8B / MBPP the best K is 2 at every step; on Dream / MBPP step 64 the best K is 3 with one small (size-26) sub-cluster. Appendix K reports top- N and clustering-method sensitivity: silhouette and significance are robust across $N \in \{10, 20, 30, 50\}$ at step 68 (the plateau peak) and across KMeans/Agglomerative/HDBSCAN. A labels-hidden $K=2$ test on the full fail+pass set yields $\text{ARI} \approx 0$, confirming the silhouette measures within-fail sub-structure rather than fail/pass

separability; supervised AUC in Appendix E reports the latter.

Cross-model comparison. LLaDA uses block-wise unmasking with block length 32, whereas Dream uses a global linear schedule. At a matched checkpoint step, the spatial distribution of unmasked tokens differs between the models, so step-axis comparisons in Figure 2 reflect approximate denoising-progress alignments rather than identical model states. We do not attempt to map between block-wise commit fractions and global mask fractions on a per-sample basis. Our AR baseline (Qwen-2.5-7B) uses last-prompt-token hidden states; this is the natural single-token analog of the DLM’s pre-generation state but is not a temporal trajectory.

Task coverage. Functional correctness is verified across four tasks (MBPP, JSON schema, GSM8K, ARC), but this still covers only a small slice of DLM use cases. Generalising the task-dependent peak finding to other open-ended tasks (instruction following, dialog, multi-turn reasoning) requires task-appropriate functional metrics and additional DLMs.

Sample size and seed dependence. Cluster sample sizes are limited by the number of fail cases per (model, dataset) cell. MBPP has ~ 140 fails per DLM, while GSM8K has fewer than 250 each. A three-seed sensitivity check on LLaDA-MBPP (Appendix I) shows pass/fail counts vary by at most ± 3 samples and the plateau structure is preserved across seeds. The temporal-oscillation phenomenon noted in concurrent DLM work (Wang et al., 2025) would primarily affect borderline samples, which our seed-stability check is consistent with.

Causal interpretation. Even with the negative steering result, our conclusion that “f15601 is a diagnostic correlate rather than a directly controllable cause under our protocol” is interpretation-dependent. A more decisive test would be a counterfactual experiment that surgically replaces the feature direction with a paired pass-cluster feature direction at the peak step, or a gradient-based attribution that traces logit differences back to the SAE layer. We leave such causal-mechanistic follow-up to future work.

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A Top-Feature Persistence Across Steps

Figure 7 shows the pairwise Jaccard of top-20 fail features between denoising steps and the enrichment trajectory of three illustrative features for LLaDA-8B / MBPP. f15601 is the only feature among the top-20 at all three peak-or-later steps (32, 64, 127), and is the top-1 fail-enriched feature at both step 32 (+0.391 enrichment) and step 64 (+0.385); on the sparse 7-grid only step 64 reaches $p < 0.05$ (Figure 1 reports the dense sweep that locates the plateau).

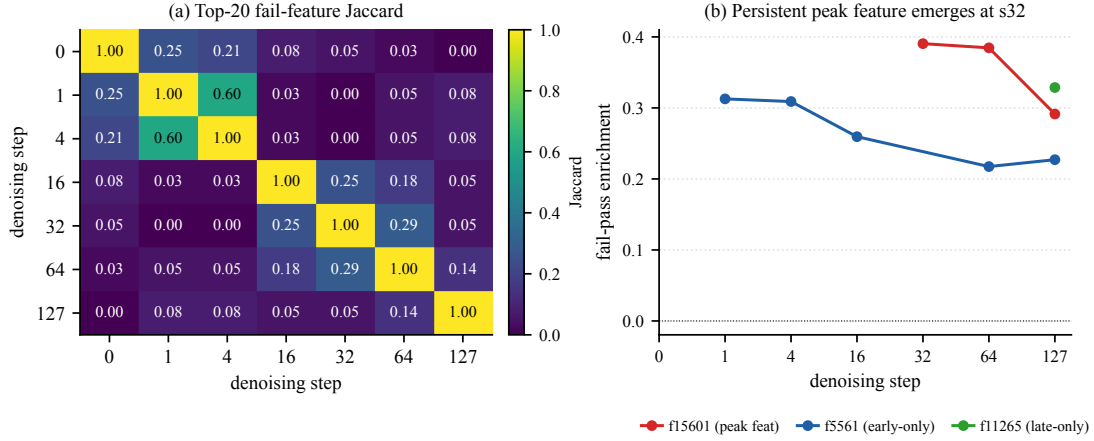


Figure 7: Left: pairwise Jaccard of top-20 fail features between denoising steps. Right: enrichment trajectory of f15601 (persistent peak), f5561 (early only), and f11265 (final only).

#	characteristic features	illustrative failures
0	f11404, f189, f8825	f9798, f496, <i>closest smaller number</i> : returns n for small inputs instead of $n-1$, failing assertions. <i>unique element in sorted array</i> : malformed pointer-update code, syntax error.
1	f189, f3320, f15601	f492, f11404, <i>remove first/last occurrence</i> : deletes only boundary characters, missing interior occurrences. <i>lowercase underscore regex</i> : uses <code>re.compile</code> without importing <code>re</code> , <code>NameError</code> .

Table 4: Illustrative generated failures from the two LLaDA-MBPP step-64 fail clusters. Error-type counts over 10 regenerated failures per cluster: cluster 0 has 3 assertion / 2 syntax / 4 name / 1 value error; cluster 1 has 5 assertion / 4 name / 1 type error.

B Qualitative Cluster Inspection

Table 4 shows illustrative LLaDA-MBPP failures from the two step-64 fail clusters. We re-generated the first ten fail samples from each cluster with the baseline LLaDA decoding setup and re-ran the MBPP tests. The examples are not intended as human-validated feature labels; they only show that both sparse clusters contain concrete functional failures, ranging from malformed code to plausible but semantically wrong implementations.

C Step-64 Cross-grid Snapshot

Table 5 summarises silhouette + permutation p across the 4 datasets $\times$ 3 models grid: DLM step-64 plateau point plus the single-step Qwen AR reference, complementing the trajectory-shape claim in Figure 2.

dataset	LLaDA	Dream	Qwen
mbpp	0.66 / 0.033	0.50 / 0.128	0.16 / 0.106
json	0.58 / 0.195	0.77 / 0.135	0.22 / 0.007
gsm8k	0.38 / 0.432	0.29 / 0.449	0.15 / 0.315
arc	0.43 / 0.312	0.18 / 0.882	0.13 / 0.343

Table 5: Silhouette / permutation p per condition: DLM step-64 for LLaDA / Dream, single-step AR reference for Qwen; bold $p < 0.05$. Full DLM trajectory in Figure 2.

D Generation and Probing Details

Denoising procedure. Both models run 128 denoising steps. LLaDA uses block-based unmasking: the generation region is divided into blocks of 32 tokens, and each block is denoised sequentially over $128/(\text{gen_length}/32)$ steps per block. Within each block, the model selects the most confident tokens to unmask at each step. Dream uses a global linear schedule: at step i , the fraction of tokens to unmask is determined by a linearly decreasing timestep schedule from $t = 1$ to $t = \epsilon$, and the most confident masked tokens are unmasked globally.

Prompt construction. For JSON schema, we prepend a system prompt containing the target schema and append a JSON code-fence marker as a generation prefix. For GSM8K, the system prompt instructs the model to solve step-by-step and end with ##### followed by the numeric answer. Both models use their respective chat templates. A complete prompt example for each task is shown in Figure 8.

Hidden state extraction. At each of the 7 checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$, we run a forward pass with `output_hidden_states=True` and extract the residual stream at the SAE layer.

The generation region is partitioned into 4 equal-length position regions and mean-pooled within each region, producing a d -dimensional vector per region (4096 for LLaDA, 3584 for Dream and Qwen). The region-mean is passed through the DLM-Scope SAE for feature analysis (§3).

Steering hook implementation. The steering hook is registered on the SAE-layer transformer block via `register_forward_hook`. The hook receives the tuple output, re-encodes `output[0]` through the SAE in-place, computes $\alpha \cdot z_{i^*} \cdot W_{\text{dec}}[:, i^*]$ and subtracts the contribution. A counter and per-step diagnostic prints confirm the hook fires $T=128$ times per generation (once per denoising step) and that the modification magnitude $\|\Delta h\|$ is non-zero.

Computational resources. All generation, encoding, and steering experiments use NVIDIA A100 (80GB) GPUs on Modal. Models are loaded in bfloat16; SAE weights cast to bfloat16 for the steering hook. Total compute: approximately 40 A100-hours across all reported experiments.

Reproducibility and determinism. Generation uses seed 0; both LLaDA and Dream commit the highest-confidence masked positions deterministically, so the fail/pass partition per cell is reproducible across runs. Hidden states are cached as bfloat16 tensors; SAE encoding reads from the dense residual cache on demand, so top-20 selection, KMeans, and permutation null are rerunnable without re-invoking either DLM. The permutation null uses 1000 shuffles per cell with a fixed permutation seed offset, so silhouette p -values are them-

selves reproducible.

E Single-feature vs. Raw-probe AUC

We compare the cross-validated AUC of (i) the single most-fail-enriched SAE feature, (ii) top- N SAE features (logistic regression with per-fold selection), and (iii) a raw-probe baseline (PCA-64 + LR on the same hidden state) across all 12 cells (Figure 9). Raw probing wins in 10 of 12 cells; the two SAE wins (Qwen / MBPP, Dream / ARC) are cells where the raw probe is itself near chance. Top-20 SAE features recover roughly 85–95% of the raw-probe AUC, suggesting sparse atoms carry most but not all of the signal in the dense residual. This complements the negative steering finding (§4): even at optimal feature selection, sparse linear features under-represent the information that determines correctness.

F Dense Temporal Sweep on MBPP

We re-generate MBPP on both DLMs and capture hidden states at 9 dense checkpoints in [48, 80] (every 4 steps) and an additional 6 checkpoints in [84, 124] (every 8 steps), then run the same TopK-SAE encoding, top-20 fail enrichment, KMeans silhouette, and permutation null pipeline as the main 7-point grid. This sweep was requested as a robustness check against sparse-sampling artefacts. Generation uses seed 0 throughout; counts are LLaDA 115/142 and Dream 125/132 pass/fail. Three seeds confirm stability (Appendix I). Figure 3 plots the signal-to-null gap per step for both DLMs across the [48, 80] plateau region; Appendix I reports the [84, 124] late-plateau extension across three seeds. Four findings emerge. **Plateau structure.** On LLaDA-MBPP the gap stays above +0.19 throughout [48, 116] with 6–9 sub-steps of 14 reaching $p < 0.05$ per seed. Aggregating the 14 plateau sub-steps via Fisher’s combined probability test gives $p = 5.99 \times 10^{-10}$ (seed 0), 9.31×10^{-8} (seed 1), and 1.22×10^{-8} (seed 2); Stouffer’s Z combination is even sharper ($Z \geq 6.09$, $p < 10^{-9}$ on all seeds). Step 64 (the only $p < 0.05$ point on the sparse 7-grid) sits in the middle of this plateau rather than at a sharp peak, so the sparse-grid claim that step 64 is the unique peak should be read as “step 64 is a representative point in the plateau”. **Late collapse.** The signal collapses sharply at step 124 (~ 3 steps before the final state), consistent with the residual stream re-organising as the generation completes block-by-block commits.

JSON schema (LLaDA chat template)

```
[BOS] system\n You are a helpful assistant
that answers in JSON. Here's the JSON schema
you must adhere to:\n <schema>\n {schema} \n
</schema>\n user\n {user input}\n assistant\n
""json\n
```

GSM8K (Dream chat template)

```
[BOS] im_start system\n Solve the math problem
step by step. End your answer with ####
followed by the final numeric answer. im_end\n
im_start user\n {question} im_end\n im_start
assistant\n
```

Figure 8: Schematic prompt examples after each model’s chat template is applied (special tokens shown as plain words for readability; the actual tokens used are model-specific control tokens). The generation region (≤ 256 masked tokens) is appended after the final assistant marker. Both models share the same system/user content; only the wrapping tokens differ.

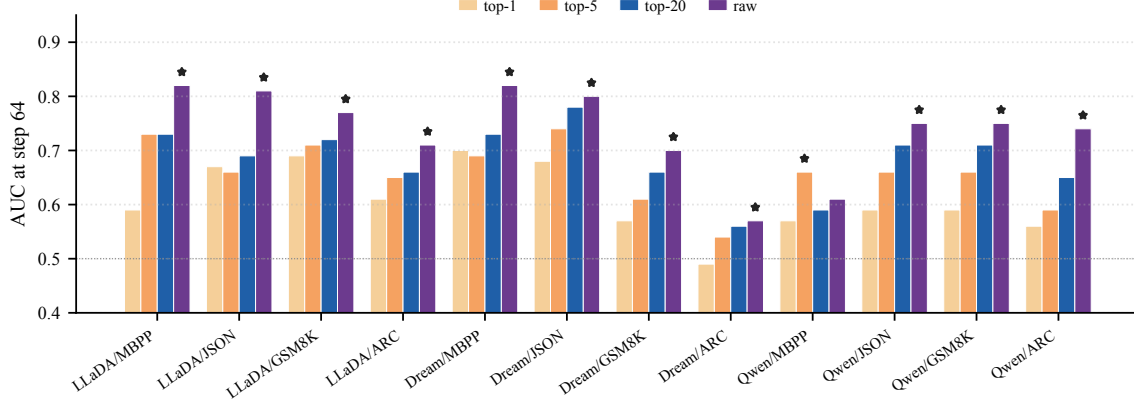


Figure 9: Per-cell 5-fold cross-validated AUC at step 64 predicting functional correctness from top- N SAE features (logistic regression with per-fold top- N selection) versus a raw-probe baseline (PCA-64 + LR on the same SAE-layer hidden state). A star marks the per-cell winner; raw probe wins in 10 of 12 cells.

Plateau feature hand-off. f15601 (LLaDA) is the top-1 fail-enriched feature at most dense steps in [48, 100] and is replaced by f3892 at the late-plateau steps 108–116. Dream-MBPP also shows a persistent top-1 marker (f6326) over [60, 76], so a similar within-plateau hand-off behaviour appears across DLMs even when the plateau itself is weaker. **Cross-DLM concentration.** Dream-MBPP shows the same temporal shape (a flat region across the middle of denoising) but at a much lower concentration: the gap never exceeds +0.16 and no sub-step reaches $p < 0.05$ under the same permutation null. The LLaDA $\times$ MBPP combination is therefore not a generic property of code generation but a model-specific point where the residual stream factors fail-vs-pass information into a small set of sparse features. This is consistent with LLaDA’s block-wise unmasking committing the function-signature block early, exposing fail-mode tokens at a denoising phase when later positions remain uncertain. We return to this contrast in §4.

G Cross-Layer SAE Specificity

To check that our choice of L26 is empirically motivated rather than an arbitrary single-layer restriction, we re-run the silhouette-with-permutation-null pipeline on cached LLaDA-MBPP plateau residuals at SAE layers {11, 16, 26, 30} (the available DLM-Scope LLaDA Mask-SAE checkpoints) at steps 64 and 68. Figure 5 shows that L26 is the unique peak: step 68 at L26 reaches gap +0.357 with permutation $p = 0.006$, while L11, L16, and L30 give gaps below +0.21 and none reach $p < 0.05$. Moreover f15601 is the top-1 fail-enriched feature only at L26 (different layers select different atoms: f13087/f541 at L11, f2741/f15654

at L16, f8020/f654 at L30), so the persistent-feature claim is itself L26-specific.

Multi-layer steering control. We further repeat the suppression protocol of §4 at the same three non-L26 layers, using each layer’s top-1 fail-enriched feature ($\alpha = 5$, $t \geq 64$) on $n = 60$ samples per layer (30 cluster-1 fails, 10 cluster-0 fails, 20 pass). All three layers yield 0/60 flips in every condition, matching the L26 within-plateau null at 0/202. The within-plateau steering null therefore generalises across the layer span; the strongest within-plateau signal (L26) is not a controllable knob and weaker layers fare no better.

H Aggregated Significance Across Cells

The Holm-Bonferroni correction on the 40 sparse-grid tests retains 0 cells (§4), but the sparse 5-step grid hides a second aggregated-significance cell. For each of the 8 (DLM, task) cells we combine the five sparse-step p -values via Fisher’s combined probability test (clamping $p_{\min} = 10^{-3}$ to avoid log 0). Figure 4 shows the result: *Dream-JSON* reaches Fisher $p = 0.047$, a second cell beyond the LLaDA-MBPP dense plateau where the within-cell trajectory exceeds chance once we look at the full trajectory rather than any single step. This corroborates the cross-DLM phase-alignment claim: code/structural tasks have aggregated significance on both DLMs (LLaDA-MBPP via dense sweep, Dream-JSON via 5-step aggregation), while reasoning tasks have aggregated significance on neither.

I Seed Sensitivity

We re-run the LLaDA-MBPP dense sweep (Appendix F) with two additional generation seeds. Pass/fail counts vary by at most ± 3 samples across

seeds (seed 0: 115/142, seed 1: 112/145, seed 2: 116/141). The plateau structure is preserved: each seed reaches $p < 0.05$ at ≥ 7 of 9 dense sub-steps in [48, 80], with f15601 as the majority top-1 fail-enriched feature across 48–80 and f3892 taking over at 108–116. The 3-seed gap-versus-step overlay is shown in Figure 1; per-seed numerical traces (gap, p -value, top-1 feature at every checkpoint) are released with the code.

J Intervention Effect Size

Across all $n=202$ samples per condition, the SAE-layer residual perturbation reaches $\|\Delta h\|/\|h\|=4.4\%$ for single-feature suppression of f15601 at s64 and s16 ($\|\Delta h\|=185$, $a_{\text{target}} \approx 9.43$), 14.8% for the top-5 combined suppression at s64 ($\|\Delta h\|=624$, $a_{\text{target}}=47.4$), and 4.4% for the reverse intervention at s64 ($\alpha = -5$). The target feature is reliably active during every intervention window, so the within-plateau null is consistent with functional correctness being distributively encoded: perturbing a sparse direction at the SAE-layer residual does not propagate into a label flip at step 64 across 202 samples, even when that perturbation reaches $\sim 15\%$ of the residual norm. Only intervention at step 16, before the plateau forms, produces any flips (3/202); the marginal effect concentrates on cluster-1 cases where f15601 fires most strongly, suggesting a weak upstream causal contribution that does not survive once the plateau features are established.

Counterfactual feature replacement. A stronger protocol simultaneously subtracts the top-5 fail-enriched directions and adds the top-5 pass-enriched directions at L26 from step 64 (replacing rather than merely ablating the fail cluster signature; top pass-enriched features at step 64 are f7961, f9732, f3011, f4420, f11861). Across the same $n = 202$ samples, this counterfactual yields 0/152 fail→pass conversions on cluster-1 and cluster-0 cases combined, and a net-zero 1 fail→pass / 1 pass→fail on the 50 pass controls. Pairing the fail-out and pass-in directions therefore does not unlock the controllability that plain suppression failed to demonstrate.

Protocol sweep. Concerned that the negative result might reflect one too-narrow protocol, we additionally sweep 17 single-layer linear configurations at L26 with $n = 15$ cluster-1 fails and $n = 5$ pass controls each: magnitude $\alpha \in \{1, 5, 10, 20, 50\}$

on f15601 at s64 (perturbation norm $\|\Delta h\|$ scales from 69 to 3152); start step $\in \{0, 4, 32, 48\}$ at $\alpha = 5$; reverse $\alpha \in \{-5, -10, -20\}$ amplifying f15601; multi-feature suppression of the top-5 and top-10 fail features; and counterfactual replacement at $K \in \{1, 5\}$ including a strong $\alpha = 20$ variant. Every one of the 17 protocols yields 0/15 fail→pass on cluster-1, for an aggregate 0/255 across the entire single-layer linear sweep on *sparse* SAE directions.

Raw-residual directional steering. The 5–15% of correctness information not captured by the top-20 sparse subspace (Figure 9) raises a natural question: would steering along a *raw* residual direction, rather than a sparse SAE feature, succeed where sparse steering fails? We test three candidate raw directions at L26 step 64: (i) the pass-minus-fail mean-difference vector ($\bar{h}_{\text{pass}} - \bar{h}_{\text{fail}}$, unit-normalised), (ii) the top principal component of the residual covariance oriented toward pass, and (iii) a Fisher LDA direction with regularised within-class covariance. Across 6 configurations (mean-diff at $\alpha \in \{5, 20, 50\}$, PCA-1 at $\alpha = 5$, Fisher at $\alpha = 5$, and mean-diff at $\alpha = 5$ from step 16), every protocol yields 0/15 fail→pass on cluster-1 fails, for an aggregate 0/90 on top of the SAE sweep. The largest perturbation ($\alpha = 50$, raw mean-diff) drives $\|\Delta h\|$ to 71,168 ($\sim 17\times$ the residual norm), well beyond the magnitudes used in successful AR steering studies, and still produces zero flips. The combined linear-steering ledger at L26 thus stands at 0/345 fail→pass over 23 protocols. This rules out the hypothesis that the sparse-subspace restriction is the culprit (raw directions occupy the full residual and still fail), and tightens the remaining explanation toward DLM-specific structural causes such as block-commit timing or non-linear computational paths. Gradient-based, multi-layer joint, and non-linear interventions remain open.

K Post-hoc Selection Sensitivity

We reanalyse cached LLaDA-MBPP plateau residuals (seed 0) at the strongest plateau steps and report three robustness checks. **(a) Top- N .** Figure 10 plots the signal-to-null gap as N varies over $\{10, 20, 30, 50\}$. Step 68 remains $p < 0.05$ at every N , while step 64 degrades for $N \geq 30$, marking step 68 as the more robust plateau point. **(b) Clustering method.** At $N=20$, KMeans, Agglomerative (Ward), and HDBSCAN (min_size=5) all recover $K=2$ with near-identical silhouette at both

steps (0.609 / 0.719 for step 64 / 68; HDBSCAN drops 11 samples as noise). **(c) Labels-hidden.** Clustering the full 257-sample fail+pass set at $K=2$ with the same top-20 features (selected via labels, cluster unsupervised) gives $\text{ARI} \approx -0.007$ at both steps ($p > 0.85$), confirming the silhouette signal measures within-fail sub-structure rather than fail/pass separability. **(d) Held-out feature selection.** To address selection-bias concerns even after the permutation null already re-selects features inside each shuffle, we partition the 142 fail samples into 5 stratified folds, select the top-20 fail-enriched features on the held-in 4 folds, and compute the KMeans silhouette on the held-out fold (mean across folds). Across the nine dense plateau steps in [48, 80], the held-out silhouette is 0.63–0.73 with held-out null mean 0.37–0.42 (gap +0.25 to +0.34); under a 500-shuffle permutation null using the same held-out pipeline, *all nine steps reach* $p < 0.05$ (best $p < 10^{-3}$ at steps 56/68/72), which is stronger than the 7/9 achieved by in-sample selection. The plateau is therefore not an artefact of label-driven feature selection.

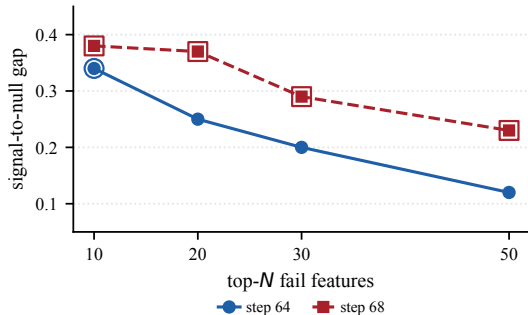


Figure 10: Signal-to-null gap as the top- N fail-feature cutoff is varied at LLaDA-MBPP plateau steps 64 and 68. Hollow rings mark $p < 0.05$. Step 68 is robust across N ; step 64 only reaches significance at $N=10$.

L Activation-Maximising Examples

To turn the statistically discovered top features into interpretable failure markers, we re-run LLaDA-8B at step 64 on all 257 MBPP samples, encode every generation position through the L26 Mask-SAE, and record the top-30 (*sample, position*) entries per target feature. Table 6 summarises the dominant context type for each feature, the fail-rate in its top-30 examples, and a representative quotation.

Two interpretive points follow. First, the mid-plateau peak feature f15601 is a *task-difficulty correlate*: its top contexts are dense mathematical specifications (Eulerian / octagonal / hexagonal / Babylonian formulae) where the model is more likely

feature	dominant context (F/P split)	top quotation
f15601	complex-math string (25/5)	doc- <i>"nth octagonal number..$O(n)=(n^2+3n)/2$"</i>
f3892	:param: return annotation (15/15)	:param tuple_list: <i>input list of tuples</i>
f5561	arithmetic variable assignment (22/8)	<i>"max_product = min_product"</i>
f11265	broken-syntax / incomplete code (17/13)	<i>"dp[n == 0] = [1 \n dp = [0]"</i>
f2144	docstring opening newline (14/16)	<i>"def fn(x):\n ""\n\n"</i>

Table 6: Activation-maximising contexts for the top fail-enriched features on LLaDA-8B / MBPP at step 64; fail/pass counts are among each feature’s top-30 examples. f15601 fires on complex-math docstring contexts (task-difficulty correlate), f11265 fires on broken-syntax contexts (closer to a true failure detector), and f2144 fires on a structural position (non-discriminative).

to fail, but the feature is not directly causal. This is consistent with the within-plateau steering null (§4): ablating a task-difficulty signal does not flip correctness because the perturbation is upstream of the failure mechanism. Second, the late-state feature f11265 fires on contexts that already contain visible syntactic defects in the partially-generated code, suggesting it is closer to a genuine failure detector than the mid-plateau atoms; the within-plateau hand-off from f15601 to f3892 (and the late shift toward f11265) thus reflects a transition from *difficulty signalling* to *outcome signalling* as denoising completes.

M Functional Correctness Definitions

Every task uses a deterministic, machine-checkable rubric; samples with no extractable candidate are labelled fail. **MBPP**: extract the first def-block or fenced “`python block, prepend test_imports,` run all assertions with a 10-second timeout; syntax errors, timeouts, and assertion failures are fails. **JSON schema**: output must parse as valid JSON and match the reference structure (top-level type, required keys, scalar-leaf types), ignoring key order and extra keys. **GSM8K**: compare only the final numeric answer; extract `####\s*([\d.]+)` and compare to gold after stripping commas and trailing zeros. **ARC-Challenge**: extract `####\s*([A-D])`; fallback to the last standalone A–D letter.